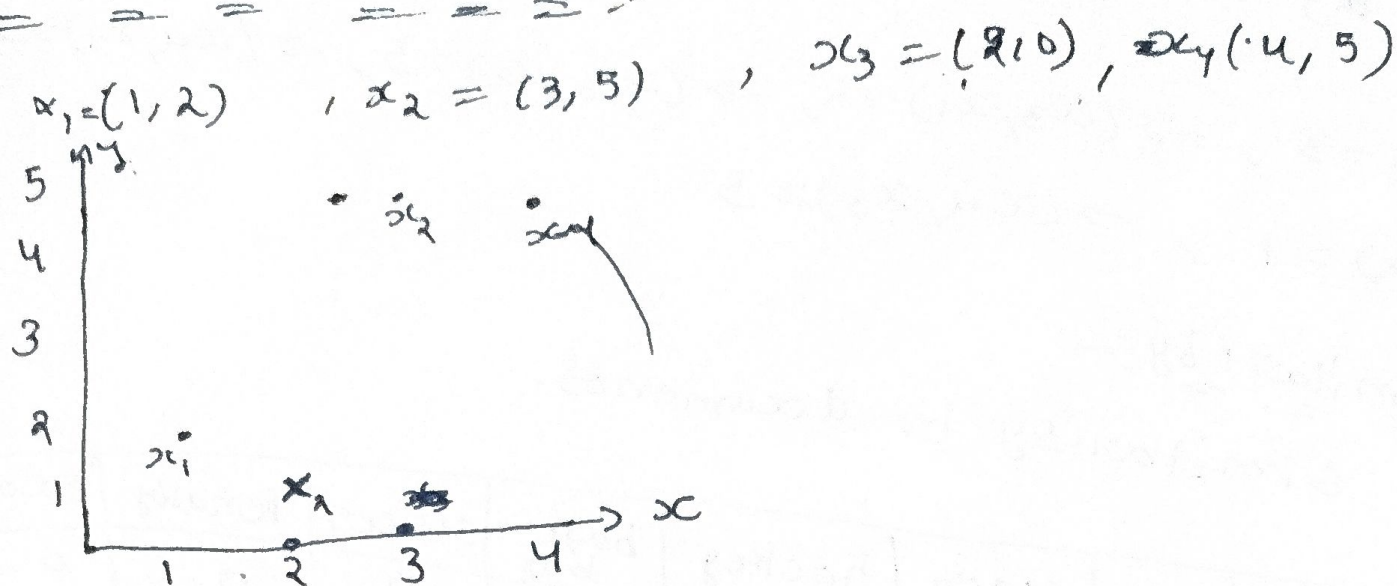


~~Activity-5~~

Activity-5

1) minowski distance :-



dissimilar, by matrices :-

→ manhattan (L_1)

L	x_1	x_2	x_3	x_4
x_1	0	-	-	-
x_2	5	0	-	-
x_3	3	6	0	-
x_4	6	1	7	0

→ $(x_2, x_1) = 5$, → $(x_3, x_1) = 3$, $(x_3, x_2) = 6$

$(x_4, x_1) = 6$, $(x_4, x_2) = 1$, $(x_4, x_3) = 7$

→ Euclidean (L_2)

L_2	x_1	x_2	x_3	x_4
x_1	0	-	-	-
x_2	3-6	0	-	-
x_3	2-2	5-1	0	-
x_4	4-25	1	5-38	0

→ $(x_2, x_1) = 36$, → $(x_3, x_1) = 2-2$, → $(x_3, x_2) = 5-1$

→ $(x_4, x_1) = 25$, → $(x_4, x_2) = 1$, → $(x_4, x_3) = 5-38$

Su permum :-

	x_1	x_2	x_3	x_4
x_1	0	-	-	-
x_2	3	0	-	-
x_3	2	5	0	-
x_4	3	4	5	0

$\rightarrow (x_2, x_1) = 3, \rightarrow (x_3, x_1) = 2, \rightarrow (x_3, x_2) = 5, \rightarrow (x_4, x_1) = 3$
 $\rightarrow (x_4, x_2) = 1, \rightarrow (x_4, x_3) = 5$

2) Cosine Similarity:-

$\rightarrow$ find similarity in documents.

Document	team	coach	hockey	base ball	soccer	penalty	score
1	5	0	3	0	2	0	0
2	3	0	3	0	1	1	0
3	0	7	0	2	1	0	0
4	0	1	0	0	1	2	2

win	loss	season
2	0	0
1	0	1
3	0	0
0	3	0

$\rightarrow$ finding similarity document 1 & 2

$$\cos(d_1, d_2) = \frac{(d_1 \cdot d_2)}{\|d_1\| \|d_2\|}$$

$\rightarrow$ where, indicates vector dot product. $\|d_1\|$ the length of the vector product.

Given

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$d_1 \cdot d_2 = 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 0 + 2 \times 1$$

$$= 15 + 6 + 2 + 2$$

$$= 25$$

$$\|d_1\| = 6.481, \quad \|d_2\| = 4.12$$

$$\cos(d_1, d_2) = \frac{25}{26.89615} = 0.94$$